

Homework assignment

Part 1

Description

Create a REST API for bank account handling in Java, [Spring Boot](#):

- Add money to account
- Debit money from account
- Get account balance
- Perform currency exchange (for simplicity with fixed exchange rates for different currencies)
- Retrieve transaction history for a given account

Requirements

- A user can have multiple accounts with separate balances
- Every account has exactly one currency (potential options are EUR, USD, SEK, GBP, VND)
- Debiting orders can only use one currency (no automatic currency exchange)
- Before debiting, simulate a call to external system (i.e. external logging) by calling some web page (like <https://httpstat.us/>)
- Microservice (self-contained)
- Saved in an SQL DB (pick your own, H2 is ok)

Part 2

Description

Create a frontend single-page application (SPA) that uses the REST API written in Part 1 using [Angular](#).

Requirements

- Home Page
 - Shows all accounts for the current user
 - Balance & currency for each account
 - Ability to click on any account to go to Account Overview Page

- Account Overview Page
 - Shows information about one specific account
 - Balance & currency for the specific account
 - Transaction history for the specific account
 - By default, only a limited number of historical transactions are loaded
 - More transactions are loaded dynamically as needed (without page reload; "infinite scroll")
 - Ability to click on any transaction to go to Transaction Overview Page
 - A [line chart](#) diagram showing the time-series data of the historic account balance (time on the horizontal axis, balance on the vertical axis)
- Transaction Overview Page
 - Shows information about one specific transaction
 - Button to export & download transaction summary as a *.pdf report

Nice to have

Make use of [NgRx](#) state management.

Note

Don't focus on making the design **pretty** - make it functional.

Submission & presentation

All code should be in a public git repository.

Be ready to do a live demo.

You will need to present your homework assignment during tech interview and address any questions.